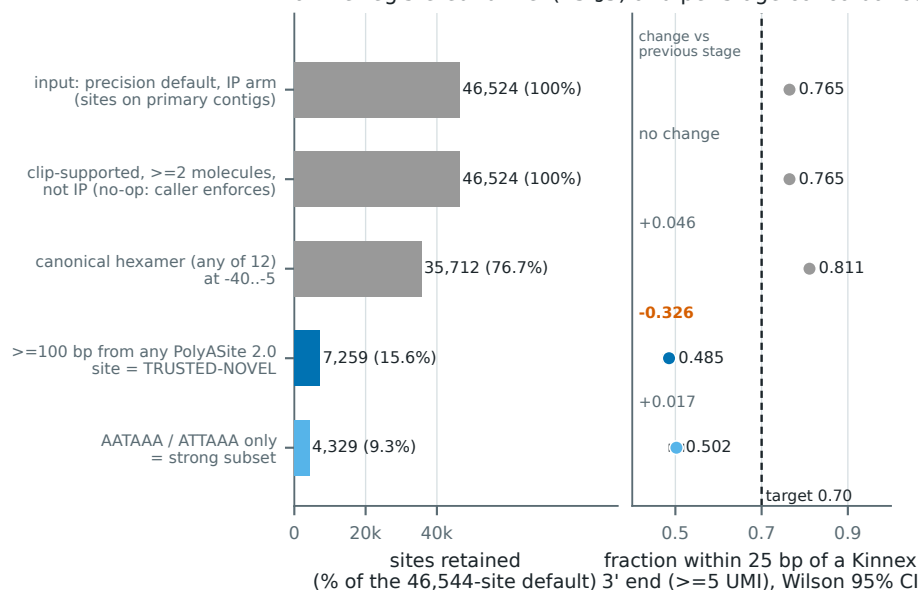


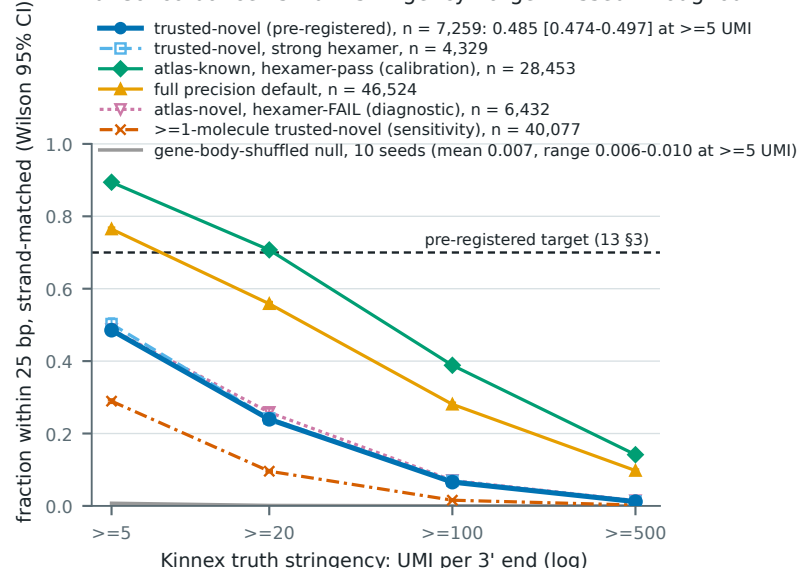
Pre-registered trusted-novel definition: 49% long-read concordance, target 70% not met

PBMC 10k v3, PeakATail precision default (IP arm, code 9dfdefb; 46,524 sites on primary contigs) -> 7,259 trusted-novel (15.6%, 3,578 genes; 4,329 with AATAAA/ATTAAG). Trusted-novel = clip-supported, >=2 molecules, not internal priming, canonical hexamer at -40...-5, >=100 bp from any PolyASite 2.0 site (13 \$3, committed before any number existed). Truth = Kinnex x3p poly(A)-verified 3' ends; hit = within 25 bp on the same strand. Result: 0.485 [0.474-0.497] at >=5 UMI (3,523 / 7,259); 0.239 / 0.066 / 0.012 at >=20 / >=100 / >=500 UMI.

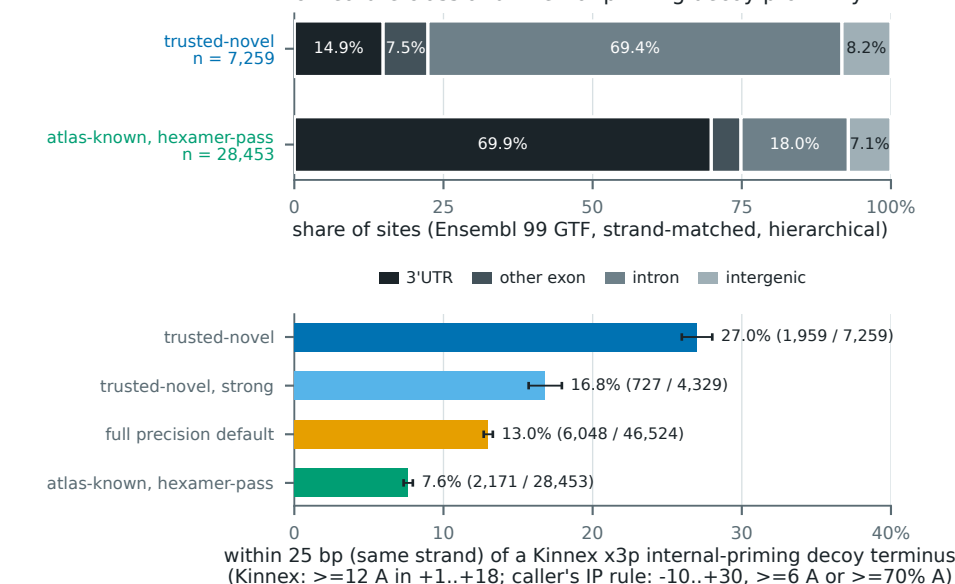
a Pre-registered funnel (13 \$3) and per-stage concordance



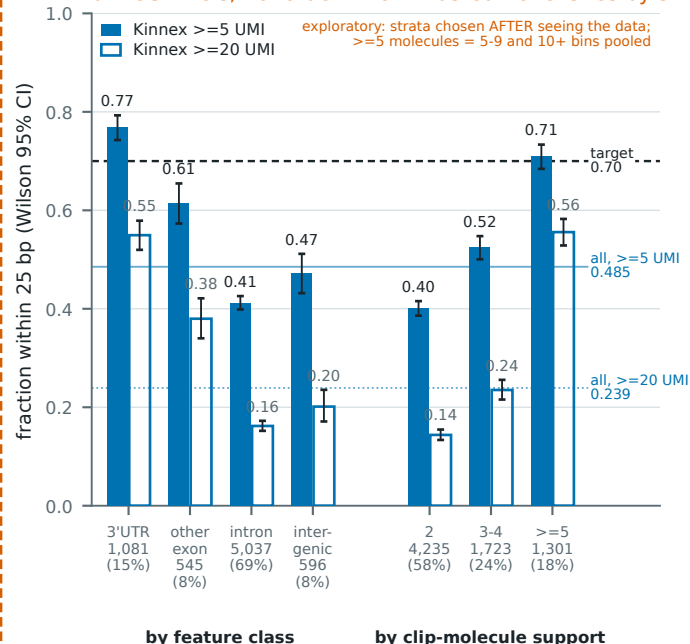
b Concordance vs truth stringency: target missed throughout



c Feature class and internal-priming decoy proximity



d POST HOC, not a definition: trusted-novel sites by stratum



Truth caveats: both Kinnex sets come from donors other than the PBMC 10k v3 donor, so a site used in this sample but absent (or <5 UMI) in the long-read donor counts as a miss -- the atlas-known hexamer-pass complement scores 0.894 under the same truth, so donor mismatch cannot explain the gap. Robustness (verifier, not part of the pre-registered metric): GEM-X truth 0.515 (3,741/7,259); x3p+GEM-X pooled 0.582 at 25 bp, 0.669 at 50 bp, 0.714 at 100 bp -- the pre-registered 25-bp metric is missed under every truth choice. Null = sites shuffled within gene bodies (10 seeds; mean 0.0074 at >=5 UMI, enrichment 65x, empirical p at the 0.091 = 1/11 floor); the sites are far from random, but the target is an absolute fraction. Funnel stages clip / >=2 molecules / not-IP remove nothing (the caller enforces them); PolyA_DB could not be used (hg19 only). The hexamer stage adds +0.046 on the whole set but nothing among atlas-novel sites (hexamer-FAIL 0.487 vs trusted-novel 0.485); the atlas-novel stage costs -0.326. Panel d is exploratory: strata chosen after seeing the data, none is a definition, none validated on held-out data (candidate: 3'UTR-restricted and/or >=5 molecules and/or '-ip-rule kinnex', Stage-1d). The v1 run (pre-fix caller, code 4efeb125) gave 0.521 on 6,628 trusted-novel sites and 24.5% decoy proximity; the corrected IP filter (#96) admits more raw peaks, so the negative result stands and slightly strengthens -- the leniency of the IP *rule*, not the strand bug, is the cause (19 \$4). Single PBMC donor, single CellRanger BAM. Sources: results/reliability/trusted_novel_final_v2_pbmc/ (REPORT_PROVISIONAL.md STATUS: VERIFIED (adversarial verifier, verdict FIXED)), manuscript/19_final_gate_v2.md \$4 (+ 16 for the v1 record). Every plotted value: results/figures/manuscript/trusted_novel_funnel*.tsv.